



QpiAI™ - Joint AI Training and Certification Program

Feature Engineering, Hyperparameter Tuning

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Recap:

**Basic Probability &
Statistics**

**Regression and
Classification**

**Linear & Logistic
Regression**

**K-Nearest Neighbor
Algorithm**

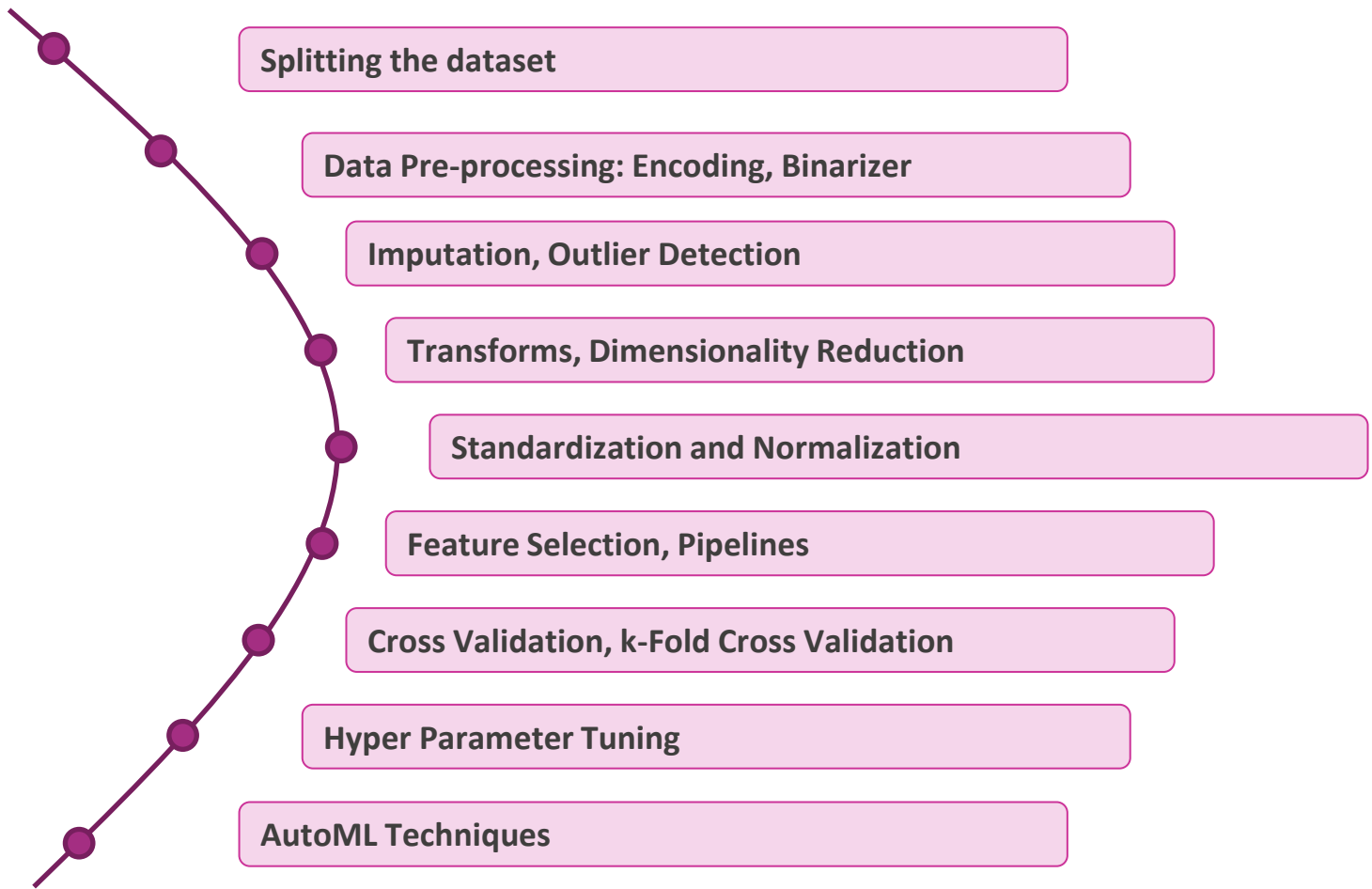
**Support Vector
Machines, Classification
Performance Metrics**

**Decision Tree,
Random Forest &
Ensemble Methods**

**Principal Component
Analysis and k- Means**

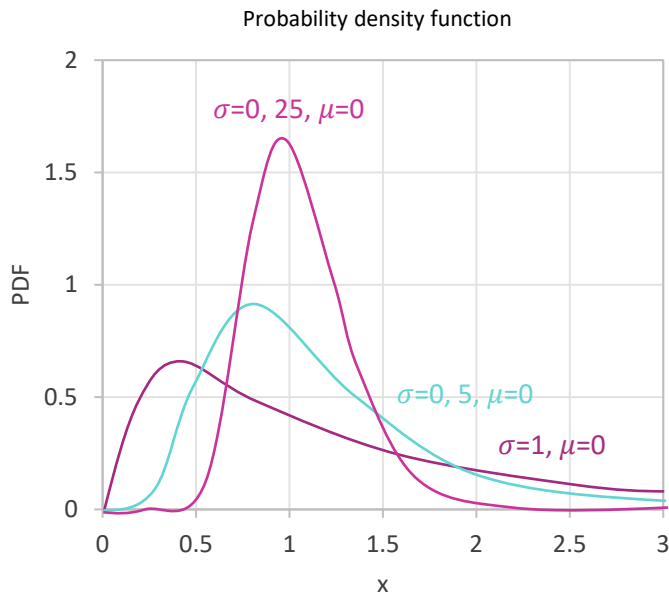
Time Series Analysis

Overview



Log-Normal Transform

- When a numerical feature does not follow the normal distribution, we can log transform this data to make it as normally distributed as possible so that the statistical analysis results from this data become more valid.
- In other words, the log transformation reduces or removes the skewness of our original data, if it approximately follows a log-normal distribution.



$$\ln(X) \sim N(\mu, \sigma^2)$$

Suppose we penalize and make θ_3, θ_4 really small

$$\min_{\theta} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2$$

$$\min_{\theta} \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + 1000 \theta_3^2 + 1000 \theta_4^2$$

Regularization Formula

$$J(\theta) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \lambda \sum_{j=1}^n \theta_j^2$$

- Whenever we try to increase the value of theta, the error term increases so we add a penalty term which forces the theta to decay towards zero
- Lambda is the regularization parameter whose value is optimized for better results

Standardization

Standardization is another scaling technique where the values are centered around the mean with a unit standard deviation by computing Z-score. This means that the mean of the attribute becomes zero and the resultant distribution has a unit standard deviation.

$$x_s = \frac{x - \mu}{\sigma}$$

Min-Max Scaling

Normalization is a scaling technique in which values are shifted and rescaled so that they end up ranging between 0 and 1. It is also known as Min-Max scaling.

$$x_N = \frac{x - x_{min}}{x_{max} - x_{min}}$$